

Publication: Reducing Health Risk Behavior and Improving Health in Adolescents With Depression

Authors

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Abstract

This randomized controlled trial evaluated the effectiveness of a brief cognitive-behavioral intervention for reducing health risk behaviors among adolescents with depressive symptoms recruited from primary care settings.

Methods

****Participants:**** A total of 350 adolescents aged 14-18 years with elevated depressive symptoms (CES-D score ≥ 16) were recruited from community health centers. Participants were excluded if they had active suicidal ideation, current substance abuse treatment, or psychotic symptoms.

****Intervention:**** Participants were randomly assigned to either a 6-session CBT intervention (n=175) or enhanced usual care (n=175). The intervention targeted both depressive symptoms and health risk behaviors including substance use and risky sexual behavior.

****Primary Outcome:**** Change in composite health risk behavior score at 6 months, measured using the Health Risk Behavior Questionnaire (HRBQ).

****Secondary Outcomes:****

****Sample Size:**** We enrolled 350 participants (175 per arm), which provided 85% power to detect a medium effect size ($d=0.40$) at $\alpha=0.05$, accounting for 20% attrition.

****Analysis:**** Intent-to-treat analysis using mixed-effects linear regression with random intercepts for participants, adjusting for baseline scores, age, gender, and site. Missing data were handled using multiple imputation ($m=20$).

Results

Of the 350 participants enrolled, 298 (85.1%) completed the 6-month assessment. The intervention group showed significant improvement in depressive symptoms (CES-D: mean difference = -3.2, 95% CI [-5.1, -1.3], $p=0.001$). However, no significant differences were found for the primary outcome of composite health risk behavior (mean difference = -0.8, 95% CI [-2.1, 0.5], $p=0.22$).

Conclusions

The CBT intervention effectively reduced depressive symptoms but did not significantly impact the primary outcome of health risk behaviors. These findings suggest that addressing depression alone may not be sufficient to change complex health risk behaviors in adolescents.